

```
Gosub getmask 'get mask for high
digit
mask1 = mask
digit = digit2
Gosub getmask 'get mask for low digit
mask2 = mask
Gosub show1 'display new mask
Gosub show2 'display new mask
WaitUs 500 'delay interval suitable
for simulation
'use large delay for the real device,
say WAITMS 500
Next i
Goto loop
End
On Interrupt 'Timer0 interrupt
routine
'continuously switch between high and
low digit displays
If phase = 0 Then
phase = 1
Gosub show1
Else
phase = 0
Gosub show2
Endif
```

```
INTCON.T0IF = 0 'enable new TMR0
interrupts
Resume
getmask: 'get appropriate 7-segment
mask for input digit
mask = LookUp(0x3f, 0x06, 0x5b, 0x4f,
0x66, 0x6d, 0x7d, 0x07, 0x7f,
0x6f), digit
Return
show1: 'show high digit on its
display
d2enable = False
PORTB = mask1
d1enable = True
Return
show2: 'show low digit on its display
d1enable = False
PORTB = mask2
d2enable = True
Return
```

Similar to the previous example, open File > Load Program and select the file 7segment.hex. Select the microcontroller PIC16F877. Go to Options > Change Clock Frequency or click on the box next to Clock Frequency in the

main window and enter the value 4.

Select 7-Segment LED Displays Panel from Tools. That will open the 7-segment displays window. In this panel, click on Setup button just below display 2. On the orange field next to Display Enable option, click twice to select the pin that will be used, that is, PORTC and 0. Click on Setup button below display 1 and click twice near Display Enable label as before, and select PORTC,1. Select the Rate to Ultimate. Start simulation by pressing F1.

Now the LED module starts displaying numbers from 0 to 99. The screenshot of this is shown in Fig. 6.

## Licence

This version of PIC Simulator IDE runs on Evaluation licence. You can start the program 30 times; each session will last for 120 minutes (two hours). After this trial period, a licence needs to be purchased for further use. ●

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